Design and Development of IoT-Based Smart Maldive Fish Processing System

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Project Proposal Report

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DECLARATION

DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

Fish processing is a deal in Sri Lanka. Drying fish is very important to make sure it is good to eat and lasts long. It is hard to control things like temperature, wetness, air movement and bad smells during drying. Now people check on the fish by hand, which can lead to mistakes and wasted fish.

Most systems that monitor the environment in agriculture and food storage only focus on temperature and wetness. They do not check for smells or air movement. We need a system that can check everything at once and send alerts if something is wrong.

This research is about creating a system that uses the internet to monitor many things in fish drying rooms. The system checks temperature, wetness, bad smells and air movement. It sends the information to a website where people can see it and make decisions.

The system uses computers, special sensors and a website to show the information. We expect this system to make fish drying more consistent reduce waste and make the fish safer to eat. This will help people who process fish and make the fish safer for people to eat.

The new system will also help make traditional industries more modern and sustainable. This means that fish will be more reliable and competitive in international markets. The system will improve food safety for consumers. Increase product reliability.

The technologies used to build the system include computers, special sensors for temperature, wetness and bad smells and a website to show the information. We use platforms like Thing Speak or Firebase to store the information and make it easy to see.

The expected results of this study are improved fish quality, reduced waste and sustainable processing methods. The system will also have a social impact by improving food safety and a commercial impact, by increasing product reliability and competitiveness.

Keywords: IoT, Environmental Monitoring, Fish Processing, Temperature and Humidity Control, Ammonia Detection

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1. INTRODUCTION

1.1 Background and Context

- **Real-World Problem**
  Fish drying is a crucial process in Maldive fish production, which is widely practiced in coastal areas of Sri Lanka. The quality and shelf life of dried fish products depend heavily on environmental conditions such as temperature, humidity, airflow, and gas levels during the drying process. In many traditional fish processing facilities, these environmental factors are controlled manually without the use of automated monitoring systems. This lack of proper environmental monitoring often leads to inconsistent drying conditions, which can result in microbial growth, product spoilage, and reduced product quality.

- **Who is Affected and Why It Is Important**
  Several stakeholders are affected by inefficient environmental monitoring in fish processing environments. Small-scale fish processors and large processing facilities may experience financial losses due to spoilage and reduced product quality. Consumers are also affected because poor processing conditions may lead to food safety risks. Additionally, exporters and businesses involved in the seafood supply chain may face challenges in maintaining product standards required for international markets. Therefore, improving environmental monitoring in fish processing is essential for ensuring product safety, maintaining consistent quality, and supporting economic sustainability in the seafood industry.

- **Statistical Evidence (Local / Global)**
  According to reports published by the Food and Agriculture Organization, post-harvest losses in fisheries can range from **20% to 30%** in many developing countries due to improper handling, storage, and environmental control during processing. These losses significantly impact food availability and economic productivity in coastal communities. In countries with strong fish processing industries such as Sri Lanka, improving processing efficiency can contribute to increased export value and reduced food waste.

- **Analysis of Existing Solutions and Their Limitations**
  Several environmental monitoring systems have been developed for agriculture and food storage industries using Internet of Things (IoT) technologies. These systems typically monitor temperature and humidity and provide basic data visualization through cloud platforms such as Thing Speak. While these systems offer useful insights, many of them focus only on a limited number of environmental parameters. They rarely integrate additional factors such as ammonia gas detection or airflow monitoring, which are critical for detecting spoilage in fish processing environments. Furthermore, most existing solutions act only as monitoring tools without providing intelligent analysis or automated alerts that can help optimize drying conditions.

- **Ideas for the Proposed Solution and Its Parts**
  To address these limitations, this research proposes an Intelligent IoT-Based Multi-Parameter Environmental Monitoring System designed specifically for fish processing environments. The proposed system integrates multiple sensors to monitor temperature, humidity, ammonia gas levels, and airflow within drying chambers. An IoT-enabled microcontroller collects sensor data and transmits it to a cloud platform for storage and analysis. A web-based dashboard allows users to view real-time environmental data and historical trends. The system also includes threshold-based alert mechanisms to notify operators when environmental conditions exceed optimal ranges, enabling timely corrective actions.

- **Relevant SDG Alignment**
  This research supports the goals of the United Nations Sustainable Development Goals (SDGs). In particular, it aligns with SDG 9: Industry, Innovation and Infrastructure, which promotes technological innovation and modernization in industrial sectors. It also supports SDG 12: Responsible Consumption and Production, which focuses on reducing food waste and improving sustainable production processes.

- **Research Cluster Alignment**
  The proposed research aligns with several interdisciplinary research clusters, including Internet of Things (IoT) and Smart Systems, Food Processing Technology, and Sustainable Industrial Innovation. By integrating sensor technologies, cloud computing, and real-time data analysis, the project contributes to the advancement of smart monitoring solutions that enhance efficiency and sustainability in traditional food processing industries.

1.2 Literature Review

- **Review of Recent Scholarly Sources**

Recent studies highlight the growing use of IoT technologies in environmental monitoring systems for food processing and aquaculture industries. Several researchers have developed IoT-based monitoring systems using sensors to measure environmental parameters such as temperature, humidity, and gas levels. For example, a study on an IoT-based fish drying device used sensors and a microcontroller to monitor temperature and humidity in drying chambers and transmit the data to a mobile application for remote monitoring. The results showed that integrating sensors with IoT platforms improved control over drying conditions and enhanced the efficiency of the drying process. (IO Informatic)

Other studies have explored environmental monitoring systems in agriculture and livestock management. Research on IoT-based monitoring in poultry environments integrated sensors such as DHT sensors for temperature and humidity and MQ-135 sensors for ammonia gas detection. The system allowed real-time monitoring through cloud platforms and helped improve environmental management efficiency. (Journal of UNM)

In addition, several IoT-enabled drying systems have been developed to optimize food processing operations. These systems monitor environmental parameters such as temperature and humidity and use cloud platforms or mobile applications for visualization and control. Research shows that IoT-enabled drying devices can significantly improve drying efficiency and product quality compared to traditional sun-drying methods that depend heavily on weather conditions. (Syntax Literate)

- **Existing Research Approaches**

Existing research mainly focuses on developing automated monitoring and control systems for food processing and aquaculture environments. Many systems use microcontrollers such as ESP8266 or ESP32 combined with environmental sensors to collect real-time data. These systems typically measure temperature, humidity, and sometimes additional parameters such as ammonia gas levels. Data collected by sensors is transmitted to cloud platforms where it can be analyzed and visualized through web dashboards or mobile applications.

Some studies also integrate advanced techniques such as fuzzy logic or intelligent algorithms to optimize environmental conditions in drying processes. These approaches allow the system to automatically adjust heating or ventilation mechanisms based on environmental conditions. Such smart monitoring systems can improve process efficiency, reduce manual intervention, and ensure more stable environmental conditions. (Zendy)

- **Existing Products and Services**

In addition to academic research, several commercial IoT solutions are available for environmental monitoring in agriculture and industrial applications. These products typically include sensor networks connected to cloud-based platforms that allow users to monitor environmental conditions remotely. Many systems provide real-time dashboards, mobile applications, and automated alerts when environmental conditions exceed predefined thresholds.

However, most of these products are designed for general environmental monitoring applications such as greenhouse monitoring, cold storage management, or livestock farming. They usually focus on monitoring temperature and humidity but rarely include integrated monitoring of additional parameters such as ammonia gas levels and airflow conditions that are particularly important in fish processing environments.

- **Limitations of Existing Systems**

Despite the progress made in IoT-based environmental monitoring systems, several limitations remain. First, many existing systems monitor only a limited number of environmental parameters, mainly temperature and humidity. Fish drying environments require monitoring of additional parameters such as ammonia gas concentration and airflow to detect early signs of spoilage and ensure efficient drying conditions.

Second, many systems function only as monitoring platforms without providing intelligent decision-support features. They collect environmental data but do not perform advanced analysis or optimization to help operators improve processing efficiency.

Third, most existing solutions are designed for general agricultural or aquaculture applications rather than specifically addressing the unique requirements of fish processing environments. As a result, there is a need for a specialized IoT-based multi-parameter monitoring system that integrates temperature, humidity, gas detection, and airflow monitoring in a single framework to optimize fish processing operations.

1.3 Research Gap

- **What is Missing in Current Research**

Although several studies have introduced Internet of Things (IoT) based environmental monitoring systems for agriculture, aquaculture, and food storage applications, limited research specifically addresses environmental monitoring in fish processing environments. Most existing studies focus primarily on monitoring temperature and humidity parameters. However, fish drying processes require monitoring additional environmental factors such as ammonia gas concentration and airflow circulation to maintain product quality and detect early signs of spoilage. The absence of a comprehensive multi-parameter monitoring framework represents a significant gap in current research.

- **Why Current Approaches Are Insufficient**

Current monitoring systems often function as basic data collection tools that simply record environmental parameters and display them through cloud platforms. While these systems provide useful information, they do not fully support intelligent decision-making or process optimization. Many existing solutions lack integrated monitoring of multiple environmental factors that directly influence fish drying efficiency. In addition, most systems do not provide automated alert mechanisms or predictive insights that can help operators take timely corrective actions. As a result, fish processing facilities may still experience inconsistent drying conditions, increased spoilage rates, and reduced product quality.

- **Novelty and Creativity of the Proposed Research**

The proposed research introduces a novel approach by developing an Intelligent IoT-Based Multi-Parameter Environmental Monitoring System specifically designed for fish processing environments. Unlike traditional monitoring systems, the proposed solution integrates temperature, humidity, ammonia gas detection, and airflow monitoring within a unified IoT framework. The system utilizes sensor networks connected to an IoT-enabled microcontroller to continuously collect environmental data and transmit it to a cloud platform for analysis and visualization. This integration of multiple environmental parameters combined with real-time monitoring and alert mechanisms provides a more comprehensive and intelligent solution for optimizing fish drying conditions.

- **Evidence-Based Justification**

Studies reported by the Food and Agriculture Organization indicate that post-harvest losses in fisheries can reach **20–30%** in developing countries due to inadequate handling, storage, and processing practices. A significant portion of these losses occurs during drying and preservation stages where environmental conditions are not properly controlled. Therefore, the development of an integrated monitoring system that can continuously track environmental conditions and provide real-time alerts is essential to reduce spoilage and improve product quality. The proposed research addresses this gap by introducing a specialized IoT-based monitoring framework tailored specifically for fish processing optimization.

Comparison of Existing Research Methods and the Proposed Method

Table 1 -Comparison of Existing Research Methods and the Proposed Method

Technologie and Methods	[1]	[2]	[3]	[4]	[5]	[6]	Proposed Method
Temperature monitoring	✓	✓	✓	✓	✗	✓	✓
Humidity monitoring	✓	✓	✓	✗	✓	✓	✓
Ammonia gas detection	✗	✗	✓	✗	✗	✓	✓
Airflow monitoring	✗	✗	✗	✓	✗	✗	✓
IoT cloud connectivity	✓	✓	✓	✓	✓	✓	✓
Real-time monitoring dashboard	✓	✓	✗	✓	✓	✗	✓
Multi-sensor integration	✗	✓	✓	✗	✓	✗	✓
Automated alert system	✗	✓	✗	✓	✗	✗	✓
Web-based monitoring system	✗	✓	✗	✗	✓	✗	✓
System designed specifically for fish processing	✗	✗	✗	✗	✗	✗	✓

Conclusion

The current literature and existing systems for environmental monitoring in fish processing highlight several limitations. Most solutions focus only on temperature and humidity, while critical factors such as ammonia gas levels, airflow, and real-time monitoring for spoilage detection are often overlooked. Furthermore, many systems act only as passive data collectors without providing intelligent alerts or decision-making support. These gaps indicate a clear need for a comprehensive, IoT-based, multi-parameter environmental monitoring system specifically tailored for fish drying processes.

By integrating multiple sensors, real-time cloud monitoring, thermal imaging, and automated alerts, the proposed research addresses these shortcomings. This solution aims to ensure consistent fish drying conditions, reduce spoilage, improve product quality, and support sustainable and modernized fish processing practices in Sri Lanka.

2. OBJECTIVE

2.1 Main Objective

The main objective of this research is to design and develop an Intelligent IoT-Based Multi-Parameter Environmental Monitoring System that continuously monitors critical environmental conditions such as temperature, humidity, ammonia gas concentration, and airflow in fish drying environments in order to improve drying efficiency, reduce spoilage risks, and enhance the overall quality and safety of processed fish products.

2.2 Specific Objectives

1. To identify and analyze the key environmental parameters (temperature, humidity, ammonia gas levels, and airflow) that influence the efficiency and quality of fish drying processes.
2. To design and implement an IoT-based sensor system using ESP32 microcontrollers and environmental sensors for real-time data collection in fish processing environments.
3. To develop a cloud-based data management system that stores environmental sensor data and allows real-time monitoring through a web-based dashboard.
4. To implement a threshold-based alert mechanism that automatically notifies users when environmental conditions exceed optimal limits for fish drying.
5. To evaluate the effectiveness of the proposed monitoring system in improving environmental stability, reducing spoilage risks, and enhancing fish processing efficiency.

3. METHODOLOGY

Research approach

This research follows a Design Science Research approach, which focuses on designing and developing a technological solution to solve a real-world problem. The study aims to design and implement an intelligent IoT-based monitoring system that can measure multiple environmental parameters in fish drying environments. The research process includes system design, prototype development, implementation, and performance evaluation. Through experimental testing, the effectiveness of the proposed monitoring system will be analyzed under real or simulated fish drying conditions.

Technical methods and algorithms

The proposed system integrates the following technical methods and algorithms:

- Multi-Sensor Data Acquisition – Environmental data will be collected using multiple sensors including temperature, humidity, ammonia gas, and airflow sensors to continuously monitor the fish drying environment.
- Signal Preprocessing and Noise Filtering – Raw sensor data will be processed using basic digital filtering techniques to remove noise and improve measurement accuracy before sending data to the cloud platform.
- Threshold-Based Monitoring Algorithm – Environmental parameters will be compared with predefined optimal threshold values to detect abnormal conditions that may affect fish drying quality.
- Real-Time Data Transmission using IoT Protocols – Sensor data will be transmitted from the ESP32 microcontroller to the cloud platform using IoT communication protocols such as HTTP or MQTT.
- Data Visualization and Trend Analysis – Collected environmental data will be visualized through graphs and charts on the monitoring dashboard to analyze trends and environmental variations over time.
- Alert Generation Algorithm – The system will generate automatic notifications when environmental parameters exceed safe limits, allowing operators to take corrective actions quickly.
- Multi-Sensor Data Integration – Data from different sensors will be combined to provide a comprehensive understanding of the drying environment and improve monitoring accuracy.
- These technical methods enable real-time environmental monitoring, early detection of abnormal conditions, and improved control of the fish drying process to ensure product quality and processing efficiency.

Tools and platforms with justification

- Microcontroller / Embedded Platform (ESP32) – Used for real-time sensor integration and data acquisition from environmental sensors such as temperature, humidity, ammonia gas, and airflow sensors. ESP32 is selected due to its built-in Wi-Fi capability and low power consumption for IoT applications.

- Environmental Sensors (DHT22, MQ-135, Airflow Sensor) – Used to measure critical environmental parameters within the fish drying environment. These sensors provide continuous monitoring of temperature, humidity, ammonia gas levels, and airflow conditions.

- Arduino IDE – Used for programming and configuring the ESP32 microcontroller to collect sensor data and transmit it to the cloud platform.

- Cloud IoT Platform (e.g., Thing Speak or Firebase) – Used for remote monitoring, real-time data visualization, and secure cloud storage of environmental sensor data.

- Web Development Technologies (Node.js and React) – Used for developing the web-based monitoring dashboard that displays environmental conditions, system alerts, and historical data trends.

- Database System (SQL / NoSQL Database) – Used to manage and store collected environmental data in a structured format for long-term monitoring and analysis.

Data Collection and Ethical Considerations

Data will be collected from experimental Maldive fish drying processes conducted under controlled environmental conditions in fish processing environments in Sri Lanka. The system will continuously monitor and record several environmental parameters that influence the efficiency and quality of the fish drying process. The following data categories will be recorded:

- Environmental data (temperature, humidity, ammonia gas levels, and airflow measurements)
- Sensor performance data (sensor accuracy, response time, and signal stability)
- System operational data (data transmission status, device connectivity, and power usage)
- Drying process data (environmental variations during different drying stages)
- Alert and threshold data (instances when environmental parameters exceed predefined limits)

Ethical considerations include:

- Ensuring that the deployment and testing of the monitoring system do not create safety risks for worker involved in fish processing operations.
- Maintaining secure storage and confidentiality of collected environmental and operational data.
- Using collected data strictly for research analysis, system validation, and performance evaluation.
- Ensuring that the research activities comply with institutional research guidelines and ethical standards.

Validation Strategy and Performance Metrics

The proposed system will be validated through experimental testing in controlled fish drying environments. The monitoring system will be deployed to collect real-time environmental data during the fish drying process. The system's performance will be evaluated by comparing sensor readings with standard environmental measurement instruments to ensure accuracy and reliability. The following performance metrics will be recorded:

•Sensor accuracy (comparison between sensor readings and reference measurement devices)
•System response time (time taken to detect and report environmental changes)
•Data transmission reliability (percentage of successful data transmission to the cloud platform)
•Alert generation efficiency (ability of the system to generate notifications when environmental parameters exceed predefined thresholds)
• System stability and uptime (continuous operation capability of the monitoring system during the drying process)

The validation process will also include repeated testing under different environmental conditions to evaluate the reliability and consistency of the system. The collected data will be analyzed to determine whether the proposed monitoring system effectively supports real-time environmental monitoring and improves the stability of fish drying conditions.

These performance metrics will be used to evaluate the effectiveness, reliability, and practical applicability of the proposed IoT-based monitoring system in fish processing environments.

TECHNOLOGIES AND PLATFORMS

Hardware Technologies

- ESP32 Microcontroller – Used as the primary embedded platform for real-time sensor data acquisition and wireless communication. It integrates multiple environmental sensors and transmits collected data to the cloud platform using WiFi connectivity. The ESP32 is selected because it provides sufficient processing capability, low power consumption, and cost-effective deployment for IoT-based monitoring systems.
- Environmental Sensors (DHT22, MQ-135, Airflow Sensor) – These sensors are used to measure critical environmental parameters within the fish drying environment. The DHT22 sensor measures temperature and humidity, the MQ-135 sensor detects ammonia gas levels associated with spoilage, and the airflow sensor monitors ventilation conditions within the drying chamber.

Software and Programming Technologies

Programming Language

- Python – Selected for backend data processing, system integration, and analytical operations. Python provides extensive libraries for IoT data handling, visualization, and system control while offering strong community support and compatibility with cloud platforms.

Embedded Programming Platform

- Arduino IDE – Used for programming and configuring the ESP32 microcontroller. It supports easy integration of environmental sensors and enables the development of firmware responsible for sensor communication and IoT data transmission.

Cloud and IoT Platforms

- Thing Speak – Used as the primary cloud IoT platform for real-time environmental data storage, monitoring, and visualization. Thing Speak enables users to remotely access sensor readings, analyze environmental trends, and generate alerts when parameters exceed predefined thresholds.

Database Systems

- MySQL / PostgreSQL – Used for storing structured environmental sensor data, system logs, configuration parameters, and historical monitoring records. These relational database systems provide data integrity, reliability, and efficient query performance for long-term environmental monitoring.
- MongoDB – Used for storing semi-structured data such as alert logs, system activity records, and sensor metadata. MongoDB offers flexible schema design and scalability, making it suitable for IoT-based data management environments.

API Testing and Integration Tools

- Postman – Used for testing and validating RESTful API endpoints used in the monitoring system. Postman ensures reliable communication between IoT devices, cloud services, databases, and the web monitoring dashboard.
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4. HIGH-LEVEL SYSTEM ARCHITECTURE

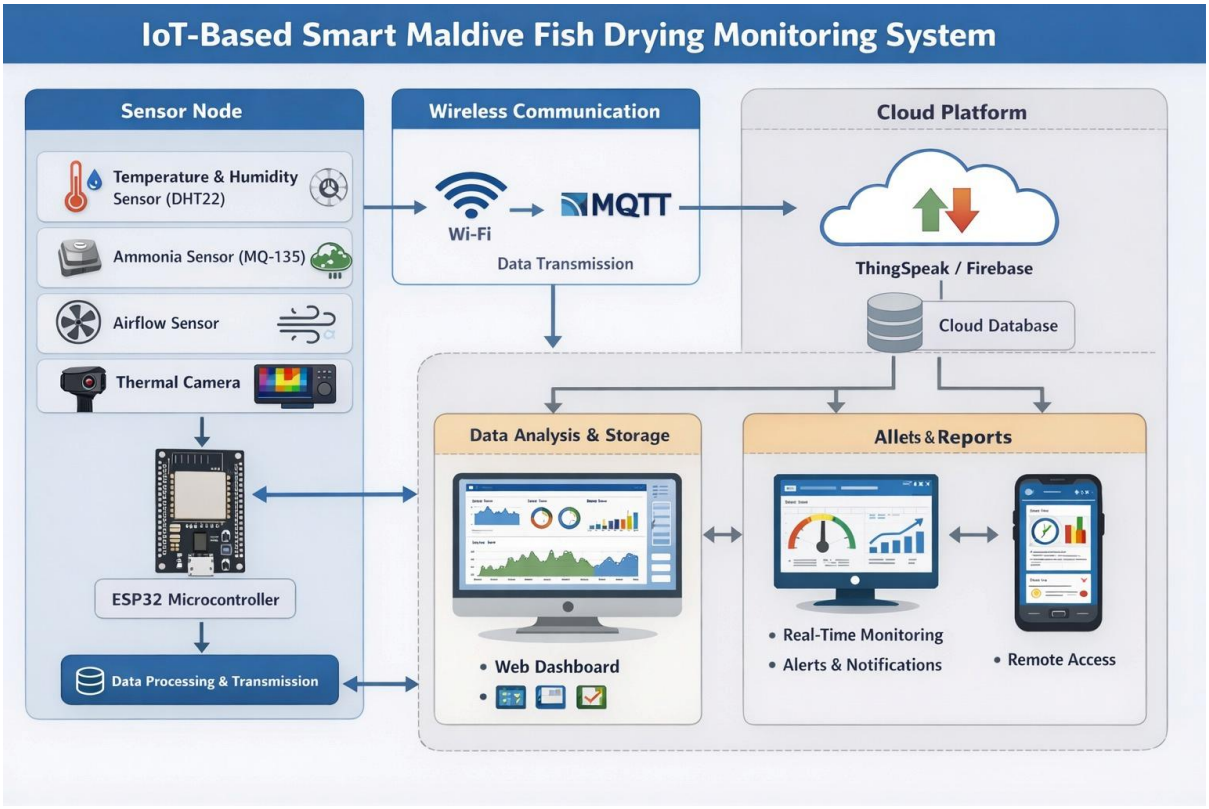


Figure 1-LEVEL SYSTEM ARCHITECTURE

5. USER REQUIREMENTS

- **Stakeholder Identification**

The proposed environmental monitoring system involves several stakeholders who interact with or benefit from the system. The primary stakeholders include fish processing facility operators who are responsible for monitoring and maintaining optimal drying conditions. These users rely on the system to observe real-time environmental parameters and respond to alerts when abnormal conditions occur. Secondary stakeholders include fish processing business owners and quality control personnel who use the collected environmental data to ensure consistent product quality and improve operational efficiency. In addition, researchers and system administrators are also stakeholders who manage system configuration, monitor performance, and analyze collected data for research and system improvement purposes. The monitoring system also indirectly benefits consumers by ensuring that processed fish products meet safety and quality standards.

- **Functional Requirements**

1. Environmental Data Monitoring – The system shall continuously monitor environmental parameters such as temperature, humidity, ammonia gas levels, and airflow in the fish drying environment.

2. Sensor Data Collection – The system shall collect real-time data from multiple environmental sensors integrated with the embedded microcontroller.

3. Wireless Data Transmission – The system shall transmit collected sensor data to a cloud platform using wireless communication.

4. Real-Time Dashboard Visualization – The system shall provide a web-based dashboard that displays environmental data in real time using graphs and charts.

5. Alert Notification Mechanism – The system shall generate alerts when environmental parameters exceed predefined threshold values.

6. Historical Data Storage – The system shall store environmental data in a database for long-term monitoring and analysis.

7. User Authentication and Access Control – The system shall allow only authorized users to access the monitoring dashboard and system data.

8. System Configuration Management – The system shall allow administrators to configure threshold values and monitoring parameters.

- **Non-Functional Requirements**

Security

The system must ensure secure communication between IoT devices and cloud platforms using secure communication protocols. Access to the monitoring dashboard must require authentication to prevent unauthorized access.

Performance

The system must process and transmit environmental data with minimal delay to enable real-time monitoring. Data transmission and dashboard updates should occur within a few seconds of sensor data acquisition.

Usability

The monitoring dashboard must provide a user-friendly interface that allows users to easily view environmental conditions, system alerts, and historical data trends.

Reliability

The system must operate continuously during the fish drying process and maintain stable data transmission without frequent interruptions.

Scalability

The system should support the addition of new sensors or monitoring nodes without significant modifications to the system architecture.

Maintainability

The system must be designed in a modular manner so that sensors, software components, and communication modules can be easily updated or replaced when necessary.

6. COMMERCIALIZATION PLAN

- **Target Market and Customer Persona**

The primary target market for the proposed system includes small and medium-scale fish processing facilities, seafood export companies, and drying plant operators in coastal regions of Sri Lanka. These businesses depend on consistent drying conditions to maintain the quality and safety of processed fish products. The typical customer persona includes fish processing plant managers, quality control officers, and business owners who require reliable environmental monitoring systems to improve operational efficiency and reduce product spoilage. Additionally, research institutions and food technology laboratories may also benefit from adopting such monitoring systems for experimental and quality assurance purposes.

- **Value Proposition**

The proposed IoT-based environmental monitoring system provides real-time monitoring of critical environmental parameters such as temperature, humidity, ammonia gas levels, and airflow within fish drying environments. By integrating multiple sensors with cloud-based monitoring and automated alert mechanisms, the system enables operators to quickly detect unfavorable environmental conditions and take corrective actions. This helps improve drying consistency, reduce spoilage rates, and enhance the overall quality of processed fish products. The system also provides data-driven insights that support better decision-making and process optimization.

- **Revenue Model**

The commercialization strategy will follow a product and service-based revenue model. The core revenue will be generated through the sale of the monitoring hardware package, which includes the IoT microcontroller, sensors, and installation setup. Additional revenue streams may include subscription-based access to advanced monitoring dashboards, cloud data storage services, and system maintenance support. Businesses may also pay for system upgrades, additional sensor modules, or customized monitoring solutions based on their processing facility requirements.

- **Cost Estimation**

The cost of developing and deploying the monitoring system includes hardware components, software development, cloud infrastructure, and installation costs. Hardware costs include the ESP32 microcontroller, environmental sensors, power supply modules, and connectivity components. Software costs involve development of the monitoring dashboard, database management systems, and cloud integration. Operational costs may also include cloud service subscriptions, system maintenance, and technical support services. Overall, the system is designed to be cost-effective so that it can be adopted by small and medium-scale fish processing businesses.

- **Pricing Strategy**

The pricing strategy will focus on affordability and scalability. The base hardware package will be offered at a competitive price suitable for small and medium-scale fish processing facilities. Additional services such as cloud analytics, system customization, and maintenance support may be offered through subscription plans or service packages. This pricing strategy ensures accessibility while maintaining sustainable revenue generation for long-term commercialization.

- **Competitive Advantage**

The proposed system offers several advantages compared to existing environmental monitoring solutions. Unlike traditional monitoring systems that focus only on temperature and humidity, the proposed system integrates multiple environmental parameters including ammonia gas detection and airflow monitoring. The integration of cloud-based monitoring, real-time alerts, and data analytics provides a more comprehensive solution tailored specifically for fish processing environments. This specialization gives the system a strong competitive advantage in the seafood processing industry.

- **Intellectual Property Considerations**

The intellectual property of the proposed system may include the system architecture design, sensor integration methods, software algorithms, and monitoring dashboard interface. Appropriate measures such as copyright protection for software components and potential patent applications for unique system designs may be considered. Protecting the intellectual property ensures that the developed technology can be commercially deployed while maintaining ownership and innovation rights.

7. BUDGET AND JUSTIFICATION

7.1 Hardware Costs

Table 2 - BUDGET AND JUSTIFICATION

Component	Quantity	Unit Cost (LKR)	Justification
ESP32 Microcontrolle	1	2000.00	Primary IoT device for sensor integration and data transmission.
DHT22 Temperature & Humidity Sensor	4	3000.00	Measures temperature and humidity in multiple locations within the drying chamber.
MQ-135 Ammonia Gas Sensor	2	2800.00	Detects ammonia gas levels to monitor fish spoilage and air quality
Thermal Camera (MLX90640)	1	15000.00	Maps temperature distribution for uniform drying
Enclosure / Housing	2	5000.00	Protects electronics from moisture and dust

8. WORK BREAKDOWN STRUCTURE (WBS)

The Work Breakdown Structure (WBS) divides the project into smaller tasks to manage design, development, testing, and documentation efficiently. The tasks are organized according to the different phases of the research project.

8.1 Task Breakdown

Table 3 - WORK BREAKDOWN STRUCTURE (WBS)

WBS ID	Task	Description
1	Literature Review	Study research papers on IoT-based fish drying systems, environmental monitoring, and multi-parameter sensors. Identify research gaps
2	Requirement Analysis	Identify system requirements including functional and non-functional requirements for the IoT-based monitoring system.
3	System Design	Design overall system architecture, workflow diagrams, and component interactions including ESP32, sensors, thermal camera, and cloud integration
4	Hardware Setup	Assemble ESP32, sensors (DHT22, MQ-135, airflow sensor, thermal camera), power supply, and connectivity modules.
5	Sensor Integration	Connect and calibrate temperature, humidity, ammonia gas, airflow, and thermal camera sensors. Ensure accurate data collection.
6	IoT & Cloud Setup	Configure Thing Speak/Firebase for real-time data transmission, storage, and visualization.
7	Web Dashboard Development	Develop a web-based dashboard for real-time environmental monitoring, graphs, heat maps, and alert notifications.
8	System Integration	Integrate all hardware, sensors, cloud platform, and dashboard to function as a single monitoring system.
9	Testing and Validation	Test the system in controlled fish drying environments. Compare sensor readings with reference instruments.
10	Performance Evaluation	Analyze system performance including sensor accuracy, response time, alert efficiency, and overall drying efficiency improvement.
11	Documentation	Prepare final report, system diagrams, user manual, and appendices including WBS, budget, and ethical considerations.
12	Final Presentation	Prepare and deliver presentation demonstrating system functionality, results, and benefits.

8.2 Timeline Alignment

The tasks are scheduled sequentially from research and design to implementation, testing, and final documentation to ensure the project is completed within the allocated timeframe.

- Early stages (Tasks 1–3) focus on planning, research, and design.
- Middle stages (Tasks 4–8) involve hardware setup, sensor integration, IoT configuration, and dashboard development.
- Later stages (Tasks 9–12) focus on testing, evaluation, documentation, and final presentation.

8.3 Workload Distribution

This project is conducted as an **individual undergraduate research project**. Therefore, all development tasks—including system design, hardware setup, sensor integration, IoT configuration, dashboard development, testing, evaluation, and documentation—are handled by the researcher with guidance from the project supervisor.

9. GANTT CHART

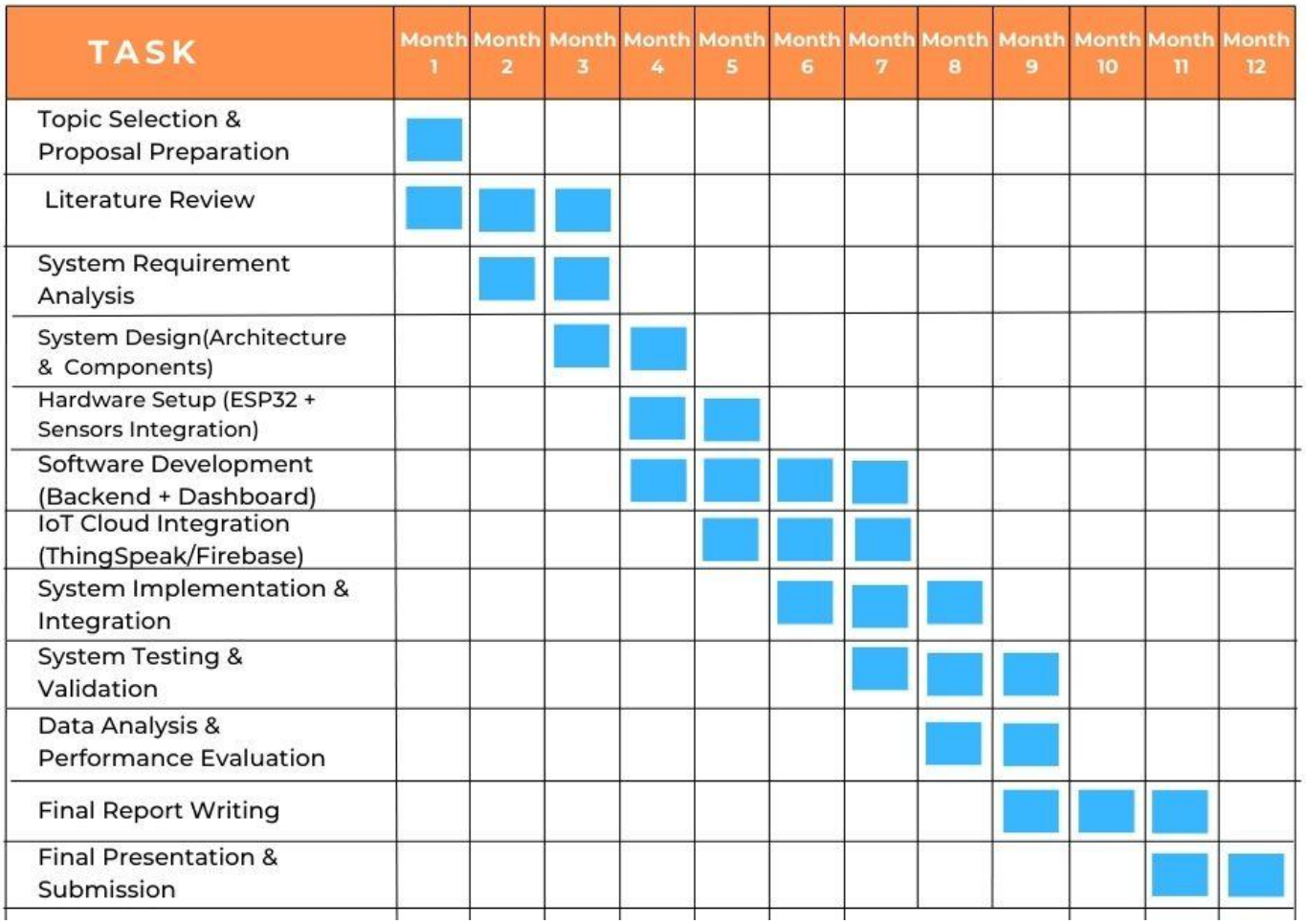


Figure 2 - Gantt Chart

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11. APPENDICES

The appendices provide supplementary materials that support the research, system design, and implementation of the IoT-Based Smart Maldive Fish System. These materials include survey instruments, diagrams, mockups, risk analysis, and ethical documentation.

11.1 Survey Instruments

- Questionnaires and interview forms used to collect data from fish processing facility operators.
- Surveys aimed at understanding current fish drying practices, challenges in environmental monitoring, and user requirements for an IoT-based system.
- Example questions include:
    - How do you currently monitor temperature and humidity in the drying process?
    - Have you experienced fish spoilage due to environmental conditions?
    - What features would you like in a monitoring system?

11.2 Detailed Diagrams

- System architecture diagrams showing the integration of ESP32, sensors (temperature, humidity, ammonia gas, airflow, thermal camera), and cloud platforms.
- Data flow diagrams illustrating sensor data acquisition, cloud transmission, storage, and visualization on the web dashboard.
- Component interaction diagrams highlighting communication between hardware, software, and user interface.

11.3 Preliminary UI Mockups

- Initial designs of the web-based monitoring dashboard.
- Visuals include:
    - Real-time graphs for temperature, humidity, and ammonia gas.
    - Thermal heat map for detecting hotspots in the drying chamber.
    - Alerts and notifications interface.

11.4 Risk Analysis

- Identification of potential risks during system design and implementation, such as:
    - Sensor malfunction or inaccurate readings.
    - IoT connectivity issues.
    - Data security and unauthorized access.

    - Safety risks during hardware setup and testing in drying environments.
- Mitigation strategies for each identified risk.

11.5 Ethical Documents

- Ethical approval forms and consent letters for any data collection from fish processing facilities.
- Confidentiality agreements to ensure secure handling of operational data.
- Compliance statements with institutional research guidelines and safety standards.